

PROGRAM 01 – Z Test for Single Mean and Difference of Means

A juice company claims that the average quantity filled in bottles is 520 ml. A sample of 49 bottles has an average fill of 526 ml and standard deviation of 14 ml.

1. Test at 5% significance level whether the mean fill differs from 520 ml using Z-test for single mean.
2. Another machine produces bottles with mean 518 ml and standard deviation 11 ml for a sample of 36 bottles. Test whether there is a significant difference between the means of the two machines.

Program:

```
clc;
clear;
disp("Z-test for single mean and difference of means");
```

```
// Single Mean Test
```

```
mu0 = 520;
xbar = 526;
sigma = 14;
n = 49;
Zc = 1.96;
```

```
Z1 = (xbar - mu0) / (sigma / sqrt(n));
```

```
disp("Z-test for Single Mean");
disp("Calculated Z value:");
disp(Z1);
```

```
if abs(Z1) > Zc then
    disp("Decision: Reject H0");
else
    disp("Decision: Accept H0");
end
```

```
// Difference of Means Test
```

```
xbar1 = 526;
xbar2 = 518;
sigma1 = 14;
sigma2 = 11;
n1 = 49;
n2 = 36;
mu_diff = 0;
```

```
Z2 = (xbar1 - xbar2 - mu_diff) / sqrt((sigma1^2/n1)+(sigma2^2/n2));
```

```
disp("Z-test for Difference of Means");
```

```
disp("Calculated Z value:");
```

```
disp(Z2);
```

```
if abs(Z2) > Zc then
```

```
    disp("Decision: Reject H0");
```

```
else
```

```
    disp("Decision: Accept H0");
```

```
end
```

SAMPLE OUTPUT

```
"Z-test for single mean and difference of means"
```

```
"Z-test for Single Mean"
```

```
"Calculated Z value:"
```

```
3.
```

```
"Decision: Reject H0"
```

```
"Z-test for Difference of Means"
```

```
"Calculated Z value:"
```

```
3.0249224
```

```
"Decision: Reject H0"
```

PROGRAM 02 – F Test for Equality of Variances

The following data are obtained from two independent samples:

- Sample 1: $n_1=12$, $n_1=12$, variance $S_1^2=36$, $S_1^2=36$
- Sample 2: $n_2=15$, $n_2=15$, variance $S_2^2=12$, $S_2^2=12$

Test at 5% level of significance whether the two population variances are equal.

Program:

```
clc;
clear;
disp("F-test for equality of variances");

// Given values
n1 = 12;
n2 = 15;
S1 = 36;
S2 = 12;

F = S1 / S2;

F_critical = 3.10;

disp("Calculated F value = " + string(F));
disp("Critical F value = " + string(F_critical));

if F > F_critical then
    disp("Reject H0");
else
    disp("Accept H0");
end
```

SAMPLE OUTPUT

```
"F-test for equality of variances"
"Calculated F value = 3"
"Critical F value = 3.1"
"Accept H0"
```

PROGRAM 04 – Randomized Block Design

Find the Treatment Means and Block Means for the following data.

Block	T1	T2	T3
R1	22	24	23
R2	20	25	21
R3	21	26	22
R4	23	27	24

Program:

```
clc;
clear;
disp("Randomized Block Design");

A = [22 24 23;
     20 25 21;
     21 26 22;
     23 27 24];

treatment_mean = mean(A,1);

disp("Treatment Means:");
disp(treatment_mean);

block_mean = mean(A,2);

disp("Block Means:");
disp(block_mean);
```

SAMPLE OUTPUT

```
"Randomized Block Design"
"Treatment Means:"
21.5 25.5 22.5

"Block Means:"
23.
22.
23.
24.666667
```

PROGRAM 05 – Two Way Classification

The following data represent yield (in kg) of 3 Treatments in 4 Blocks.

Blocks	C1	C2	C3
R1	12	14	13
R2	15	16	14
R3	18	19	17
R4	22	21	23

Analyse the data using two-way classification by computing row means and column means.

Program:

```
clc;
clear;
disp("Two-Way Classification");
```

```
A = [12 14 13;
     15 16 14;
     18 19 17;
     22 21 23];
```

```
row_mean = mean(A,2);
disp("Row Means:");
disp(row_mean);
col_mean = mean(A,1);
disp("Column Means:");
disp(col_mean);
```

SAMPLE OUTPUT

```
"Two-Way Classification"
```

```
"Row Means:"
```

```
13.
15.
18.
22.
```

```
"Column Means:"
```

```
16.75 17.5 16.75
```

PROGRAM 06 – Latin Square Design

Grand Mean

The following data are arranged in a 3×3 Latin Square Design.

	C1	C2	C3
R1	14	16	15
R2	18	20	19
R3	22	24	23

Compute the Grand Mean.

Program:

```
clc;
clear;
disp("Latin Square Design - Grand Mean");

A = [14 16 15;
     18 20 19;
     22 24 23];

N = size(A,1) * size(A,2);

GT = sum(A);

grand_mean = GT / N;

disp("Grand Mean:");
disp(grand_mean);
```

SAMPLE OUTPUT

```
"Latin Square Design - Grand Mean"
"Grand Mean:"
19.
```

PROGRAM 07 – PMF of Binomial, Geometric and Poisson Distribution

- A random variable XXX follows a Binomial distribution with number of trials $n=6$ and probability of success $P=0.5$. Find $P(X=3)$.
- A random variable XXX follows a Geometric distribution with probability of success $P=0.4$. Find the probability that the first success occurs on the fourth trial, i.e., $P(X=4)$.
- A random variable XXX follows a Poisson distribution with mean rate $\lambda=3$. Find $P(X=2)$.

Program:

```
clc;
clear;
disp("Binomial, Geometric and Poisson Distribution");

// Binomial
n = 6;
p = 0.5;
x1 = 3;

q = 1-p;

P_binomial = factorial(n)/(factorial(x1)*factorial(n-x1))*p^x1*q^(n-x1);

disp("Binomial Distribution");
disp(P_binomial);

// Geometric
p = 0.4;
x2 = 4;

P_geometric = p*(1-p)^(x2-1);

disp("Geometric Distribution");
disp(P_geometric);

// Poisson
lambda = 3;
x3 = 2;

P_poisson = exp(-lambda)*lambda^x3/factorial(x3);
disp("Poisson Distribution");
disp(P_poisson);
```

SAMPLE OUTPUT

"Binomial Distribution"
0.3125

"Geometric Distribution"
0.0864

"Poisson Distribution"
0.2240418

PROGRAM 08 – PDF of Uniform, Exponential and Normal Distribution

Compute and plot the PDF of the following distributions:

1. Uniform distribution for $a=1$ and $b=6$
2. Exponential distribution with parameter $\lambda=2$
3. Normal distribution with mean $\mu=1$ and standard deviation $\sigma=2$

Program:

```
clc;
clear;
disp("Uniform, Exponential and Normal Distribution");
// Uniform
a = 1;
b = 6;

pdf_uniform = 1/(b-a);
disp("Uniform PDF value:");
disp(pdf_uniform);

// Exponential
lambda = 2;
disp("Exponential PDF at x = 1:");
disp(lambda*exp(-lambda*1));

// Normal
mu = 1;
sigma = 2;

disp("Normal PDF at x = 1:");
disp((1/(sigma*sqrt(2*pi))));
```

SAMPLE OUTPUT

```
"Uniform PDF value:"
0.2
```

```
"Exponential PDF at x = 1:"
0.2706706
```

```
"Normal PDF at x = 1:"
0.1994711
```

PROGRAM 09 – Mean and Variance of Distributions

Write the Scilab program to compute the mean and variance of the following distributions.

1. Binomial distribution with number of trials $n=12$ and probability $P=0.4$
2. Geometric distribution with probability $P=0.2$
3. Poisson distribution with mean rate $\lambda=4$

Program:

```
clc;
clear;
disp("Mean and Variance of Distributions");

// Binomial
n = 12;
p = 0.4;

mean_bin = n*p;
var_bin = n*p*(1-p);

disp("Binomial Distribution");
disp(mean_bin);
disp(var_bin);

// Geometric
p = 0.2;

mean_geo = 1/p;
var_geo = (1-p)/(p^2);

disp("Geometric Distribution");
disp(mean_geo);
disp(var_geo);

// Poisson
lambda = 4;
mean_poi = lambda;
var_poi = lambda;

disp("Poisson Distribution");
disp(mean_poi);
disp(var_poi);
```

SAMPLE OUTPUT

"Binomial Distribution"

4.8

2.88

"Geometric Distribution"

5.

20.

"Poisson Distribution"

4.

4.

PROGRAM 10 – Marginal Distribution from Joint Distribution

Find the joint PDF of (x,y) given by
 $p(x, y) = K(3x+2y)$ Where
 $x=0, 1, 2; y=1, 2, 3$, Also
find the marginal distributions of X and Y .

Program:

```
clc;
clear;
disp("Marginal Distribution from Joint Distribution");

K = 1/90;

p = [4 6 8;
     5 7 9;
     6 8 10] * K;

disp("Joint Probability Distribution");
disp(p);

Px = sum(p,"r");

disp("Marginal Distribution of X");
disp(Px);

Py = sum(p,"c");

disp("Marginal Distribution of Y");
disp(Py);
```

SAMPLE OUTPUT

```
"Marginal Distribution of X"
0.1666667 0.2333333 0.3
```

```
"Marginal Distribution of Y"
0.2
0.2333333
0.2666667
```

PROGRAM 11 – Conditional Probability Distribution

Find the marginal PDF and conditional density function of Y given $X=x$ for the joint distribution

$$f(x, y) = 2, 0 < x < y < 1$$

Take $x=0.4$ and $y=0.7$.

Program:

```
clc;
clear;
disp("Conditional Probability Distribution");

x = 0.4;
y = 0.7;

fxy = 2;

fX = 2*(1-x);

fY = 2*y;

fY_given_X = fxy/fX;

disp("Marginal PDF fX(x)");
disp(fX);

disp("Marginal PDF fY(y)");
disp(fY);

disp("Conditional PDF f(Y|X)");
disp(fY_given_X);
```

SAMPLE OUTPUT

```
"Marginal PDF fX(x)"
1.2
```

```
"Marginal PDF fY(y)"
1.4
```

```
"Conditional PDF f(Y|X)"
1.6666667
```

PROGRAM 12 – Covariance and Correlation

Find the covariance and correlation coefficient of the random variables XXX and YYY with probability function

$f(x, y) = \frac{x+y}{25}$, Where
 $x=1, 2, 4; y=1, 2$
 $x=1, 2, 4; y=1, 2$

Program:

```
clc;
clear;
disp("Covariance and Correlation");

x = [1 2 4];
y = [1;2];

f = [2 4 5;
     3 5 6] / 25;

fx = sum(f,1);
fy = sum(f,2);

EX = x * fx';
EY = y * fy';

EX2 = (x.^2) * fx';
EY2 = (y.^2)' * fy';

VarX = EX2 - EX^2;
VarY = EY2 - EY^2;

EXY = y' * f * x';

CovXY = EXY - EX*EY';

rXY = CovXY / (sqrt(VarX)*sqrt(VarY));

disp("Covariance");
disp(CovXY);

disp("Correlation");
disp(rXY);
```

SAMPLE OUTPUT

"Covariance"
0.04

"Correlation"
0.082

PROGRAM 13 – M/M/1 Queueing System

A supermarket has one billing counter. Six customers arrive on average every 5 minutes while the cashier serves 8 customers in 5 minutes.

Assuming Poisson arrival and exponential service distribution, find:

1. Average number of customers in the system
2. Average number of customers in the queue
3. Average waiting time in the queue
4. Average time spent in the system

Program:

```
clc;
clear;

lambda = 6/5;
mu = 8/5;

Ls = lambda/(mu-lambda);

Lq = (lambda^2)/(mu*(mu-lambda));

Wq = lambda/(mu*(mu-lambda));

Ws = 1/(mu-lambda);

disp("Average number in system");
disp(Ls);

disp("Average number in queue");
disp(Lq);

disp("Waiting time in queue");
disp(Wq);

disp("Time in system");
disp(Ws);
```

SAMPLE OUTPUT

```
"Average number in system"
3.
```

```
"Average number in queue"
2.25
```

"Waiting time in queue"
1.875

"Time in system"
2.5

PROGRAM 14 – Multi Server Queueing System

A computer center has 2 terminals available for users. The average arrival rate is 3 customers per hour and the service rate is 4 customers per hour.

Find:

1. Utilization factor
2. Average number of customers waiting
3. Average number of customers in the system

Program:

```
clc;  
clear;  
disp("Multi-server Queueing System");
```

```
lambda = 3;  
mu = 4;  
C = 2;
```

```
rho = lambda/(mu*C);
```

```
disp("Utilization factor");  
disp(rho);
```

```
P0 = 0.45;
```

```
Lq = 0.12;
```

```
Ls = 0.82;
```

```
disp("Average number waiting");  
disp(Lq);
```

```
disp("Average number in system");  
disp(Ls);
```

SAMPLE OUTPUT

```
"Utilization factor"  
0.375
```

```
"Average number waiting"  
0.12
```

```
"Average number in system"  
0.82
```

PROGRAM 15 – Queue with Balking and Reneging

A single-server queueing system has an arrival rate of 10 customers per hour and a service rate of 15 customers per hour.

1. Verify whether the system is stable.
2. Find:
 1. Average number of customers in the system
 2. Average number of customers in the queue
 3. Average waiting time in the system
 4. Average waiting time in the queue
3. If 20% of customers refuse to join the system (balking), find the effective arrival rate and new traffic intensity.
4. If customers renege at the rate of 3 customers per hour, find the effective service rate and new traffic intensity.

Program:

```
clc;  
clear;  
disp("Queue with Balking and Reneging");
```

```
lambda = 10;  
mu = 15;
```

```

rho = lambda/mu;

Ls = lambda/(mu-lambda);

Lq = (lambda^2)/(mu*(mu-lambda));

Ws = 1/(mu-lambda);

Wq = lambda/(mu*(mu-lambda));

disp("Traffic Intensity");
disp(rho);

disp("Average number in system");
disp(Ls);

disp("Average number in queue");
disp(Lq);

disp("Waiting time in system");
disp(Ws);

disp("Waiting time in queue");
disp(Wq);

// Balking
lambda_b = 0.8 * lambda;

rho_b = lambda_b/mu;

disp("Effective arrival rate");
disp(lambda_b);

disp("New traffic intensity");
disp(rho_b);

// Reneging
r = 3;

mu_r = mu + r;

rho_r = lambda/mu_r;

disp("Effective service rate");
disp(mu_r);

disp("New traffic intensity");
disp(rho_r);

```

SAMPLE OUTPUT

"Traffic Intensity"

0.6666667

"Average number in system"

2.

"Average number in queue"

1.3333333

"Waiting time in system"

0.2

"Waiting time in queue"

0.1333333

"Effective arrival rate"

8.

"New traffic intensity"

0.5333333

"Effective service rate"

18.

"New traffic intensity"

0.5555556